SwasthyaTwin AI — Multisignal Health Intelligence

Happiest Health Digital Twin Challenge 2026 | Team: 24cb012

- Team Name: 24cb012
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- Institution: KPR Institute of Engineering and Technology, Coimbatore, Tamil Nadu
- Department: B.Tech Computer Science and Business Systems
- Vision: Understand today. Prepare for tomorrow.
- Clinical Status: Proof-of-Concept Research Prototype (Non-Diagnostic Software)

The Healthcare Challenge & Target Users

The Failure of Generic Population Averages

- Population Norm Bias: Legacy apps compare everyone to 60-100 bpm population norms.
- Missed Drift: An athlete (resting HR 46 bpm) drifting to 62 bpm is marked 'Normal'.
- Opaque Scoring: Proprietary black-box sleep and readiness scores cannot be audited.
- Target User 1: Proactive individuals seeking early personal baseline deviation detection.
- Target User 2: Preventive clinicians needing transparent mathematical audit trails.

Why a Digital Twin Approach?

Self-Baseline Grounding & Prospective In-Silico Simulation

- Continuous Biological Modeling: Calibrated dynamically to an individual's personal history.
- Anti-Population Norms: Uses normalized median absolute deviation (nMAD) for robust corridors.
- Counterfactual 'What-If' Sandbox: Models lifestyle interventions (e.g. +1.5h sleep) in-silico.
- Data Isolation Guarantee: Simulations write to db.simulations and NEVER mutate measured history.
- 100% Explainability: Universal 'How Calculated' modals expose formulas and arithmetic.

Distinctive Innovations of SwasthyaTwin AI

What Sets This Platform Apart

- Autonomous Spatial Twin: Dark spatial canvas with 6 organ subsystems and hotkeys (1-6).
- Schematic Vector Topology: Explicit locator notice prevents CT/MRI diagnostic confusion.
- 5-Tier Data Provenance: MEASURED, DERIVED, PREDICTED, SIMULATED, SYNTHETIC.
- Universal Mathematical Modals: Step-by-step arithmetic matching displayed telemetry.
- Zero-Crash Resilience: Automatic mongomock fallback for offline demonstration.

Technical Architecture & Data Flow

Modern Full-Stack Engineering with Zero Shortcuts

- Frontend: React 19, TypeScript, Vite v8.3.4, Tailwind CSS, Recharts (Build: 2.08s).
- Backend: Python 3.13, FastAPI, Pydantic v2 (Strict canonical units: bpm, mmHg, hrs).
- Database: MongoDB 7.0 with compound indexing on patient_id and descending timestamps.
- AI Plausibility Guard: Rejects impossible vitals (HR=700 bpm, SBP < DBP) before storage.
- Multi-Tenant Isolation: Mandatory {'patient_id': id} scoping blocks cross-tenant leakage.

Machine Learning Contribution & Real Evaluation

PhysioNet BiDSleep 5-Stage AASM Sleep Model

- Real Dataset: PhysioNet BiDSleep v1.0.1 (31 adult participants, 15,765 30s epochs).
- Disjoint Group Partitioning: Bidslab00 (Train), Bidslab01 (Val), Bidslab02 (Test).
- Zero Participant & Preprocessing Leakage: RobustScaler fit strictly on training set.
- Held-Out Subject Bidslab02 (5,280 epochs / 44 hrs): Macro F1 28.22% (+18.69% lift).
- Balanced Accuracy: 34.70% (+14.70% lift) | Sensor dropout (<2.0 samples) -> Unknown.

3-Minute Live Demonstration Workflow

Repeatable End-to-End Walkthrough

- Step 1: Flagship Spatial Twin — Press 1-6 for instant organ telemetry & baseline corridors.
- Step 2: Patient Switching — Switch to PT-1001 (Aarav Mehta) or PT-1003 (48 bpm athlete).
- Step 3: Plausibility Guard — Input 999 bpm -> instant rejection; input 74 bpm -> persisted.
- Step 4: Explainability Modal — Inspect Modified Z derivation and Vitality deductions.
- Step 5: Sleep Research Workspace — 120-epoch dual hypnogram (Dreem 2 EEG vs PPG model).
- Step 6: What-If Simulator — Run +1.5h sleep scenario; show recovery curve in isolated db.

Safety, Limitations & Future Roadmap

Responsible AI in Healthcare

- Clinical Boundary: Non-diagnostic prototype; does not diagnose arrhythmias or apnea.
- Zero Imputation Policy: Missing sensor intervals are NEVER filled with synthetic zeros.
- History Gating: Minimum 14 observations enforced before establishing statistical corridor.
- Verification Evidence: 282 automated tests passed (227 Pytest + 55 Vitest, 0 failures).
- Next Steps: Web Bluetooth Low Energy (BLE) packet ingest & SMART-on-FHIR EHR sync.